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FEATURE EXTRACTION IN DEVELOPMENT OF BRAIN-COMPUTER INTERFACE: A CASE STUDY

Mark Polak[1,2], *IEEE Student Member*, and Aleksandar Kostov[2,3], *IEEE Member*

[1]Department of Computing Science, University of Alberta
[2]Faculty of Rehabilitation Medicine, University of Alberta
[3]Glenrose Rehabilitation Hospital, Edmonton, AB, Canada

Abstract -- **In this study we compared two spectral estimation methods for feature extraction in development of computer access for communication and control based on electroencephalographic (EEG) signals. We compared power spectrum calculated by fast Fourier transform (FFT) and auto regressive (AR) coefficients calculated by Burg's algorithm. We analyzed four surface EEG signals recorded above sensory-motor areas, while the subject was attempting to use only mental activities to modulate his EEG signals resulting in desired simple movements of animated object on the feedback computer screen. To establish communication channel based on EEG signals, we asked our subject to determine which mental activity produced reproducible control actions. During the on-line training we used AR coefficients of recorded EEG signals for the feedback generating variables, i.e. the object's movement direction and speed were dependent on the EEG pattern. Our subject was able to determine which mental activity resulted in desired movements and to learn to control object movements with more than 90% accuracy. We performed off-line analysis to determine if FFT or AR can be more successful in extracting relevant information from the EEG signals before performing pattern recognition by a machine learning classifier. Probability of misclassification was used as the outcome measure. Off-line analysis showed that the overall best classification accuracy achieved with AR exceeds that achieved with FFT.**

I. INTRODUCTION

Current assistive devices for people with severe disabilities rely on the user to have some neuromuscular control. In case of extreme disabilities a person may be locked into their bodies without the ability to answer even simple yes or no questions. Without any physical channels for communication, the only alternative for these people may be in exploring indirect voluntary modulation of electrical fields resulting from neural processes in their brains.

It has been shown that most subjecs can with training learn to manipulate the power distribution in their EEG-spectra [1]. The advances in computing hardware have made it possible to explore development of a real-time brain-computer interface. Our efforts are directed toward development of such a device which will extract the user's intentions quickly and with high accuracy. What makes this goal challenging is the poor resolution and reliability of the information detectable in the EEG. The information in a surface EEG recording is severely limited by the vast number of electrically active neuronal elements, the complex geometry of the brain and head, and the trial-to-trial variability in brain operations. This is a difficult asynchronous signal detection problem because the transient voluntary EEG modulated (VEM) potentials reside in non-stationary background EEG activity with very low signal-to-noise ratios (typically below -5dB [2]). To overcome these challenges, powerful feature extraction and classification methods are required. The purpose of feature extraction is to reduce the data by measuring certain features that capture the relevant information. In general, feature extraction is much more problem dependent than classification. In this study we focus on feature extraction while using a non-linear classifier.

The classical approach to feature extraction in BCI is to estimate the power at carefully chosen frequency bands in a FFT generated spectra [3,4,5,6]. Although AR spectral analysis has been studied in EEG processing for a long time [7,8,9] it has only recently been actively applied in BCI systems [10,11,12,13]. AR has two important advantages over FFT with respect to BCI. The first advantage is that AR modeling results in better frequency resolution. The AR estimation is a function of continuously varying frequency so there is no special significance to specific equally spaced frequencies as there is in FFT. The AR estimate may have very sharp spectral features so one can evaluate it on a very fine mesh near to those features, and more coarsely farther away from them [13,14,15]. The second advantage is that with AR, good spectral estimates can be obtained from short EEG segments [12]. Even very slow waveforms with a wevelength longer than the interval observed can be detected accurately [13]. In a BCI system, shortening the required time segment is important for reducing the delay in the feedback provided to the user.

The most serious disadvantage in AR spectral estimation is the problem of selecting the proper model order. With noisy input signals, if the order is too high, the result will be spurious peaks in the spectra [15]. However, too low an order yields rather smooth spectra. Some experts recommend the use of the AR method in conjunction with more conservative methods, such as periodograms, to help choose the correct model order, and to avoid getting fooled by spurious spectral features [14].

In terms of operations count, calculating the AR coefficients using the Burg's method is in the order N times M, where N is the size of the signal and M is the model order, as compared to NlogN for the FFT [14]. Therefore AR will become slightly slower than FFT if M is greater than logN, which is the case in real-time BCI systems where short signal segments are used.

An alternative to using specific frequency bands calculated by AR or FFT as features, is to use the AR model coefficients alone in a classification procedure [13], as was

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done in this study. The AR coefficients contain most of the information of the signal that would be in the spectrum and therefore a classifier should be able to discriminate between sets of AR coefficients calculated from signals with different spectral properties. The advantage of classifying with raw AR coefficients is that one does not need to search for specific frequency components that contain the information. The purpose of this work is to explore the better method of feature extraction for our BCI system between FFT spectra and raw AR coefficients, without using any prior knowledge about the spectral features.

II. METHODS

A. Experimental Setup

The subject is comfortably seated in front of a monitor while EEG signals are recorded using an electrode cap with 28 gel filled electrodes distributed according to the 10-20 international electrode system [17], one ground electrode and the linked ears reference. The electrode cap and EEG-preamplifiers are optically isolated from the rest of the equipment. This provides safety for both the subject and the operator. For signal amplification and filtering we use the Brain Imager (Neuroscience Inc.). Analog EEG signals are then digitized at 200 samples/s by a data acquisition card (DAQ) inserted in an IBM PC compatible computer. The same computer has special video card splitting the video output into two high resolution monitors, one for the subject and one for the operator. The subject uses two manual switches to mark sequences of voluntary attempts to mentally control the movement of a circular object on the feedback screen. Since mental concentration is required to produce desired EEG signals, these switches allow the subject to rest during the experiment and avoid fatigue. The goal is to move the object on the screen to a target. The position of the target is switched between UP and DOWN at the end of each run when the object reaches the target or the opposite end of the screen is hit.

B. Subject Training

We used one subject in this study who acquired reasonable control over the object on the screen. Our subject was able to achieve some control in his second training session and had a total of 14 prior sessions before the EEG experiment that is presented here. Each of the sessions lasts approximately 30 minutes. The first half of each session is used to train a new classifier and the second half is used to evaluate the performance. Performance is evaluated in terms of how many times the target is hit versus missed at various movement speed of the object. During the sessions, position of the object is updated every 50 milliseconds. In the training and evaluation of each subject, AR coefficients are used as features to the real-time classifier. In the session for which the EEG data is analyzed in this study, the subject was able to hit the target 90% or better at all object speeds where the minimum time to hit or miss was 130 milliseconds.

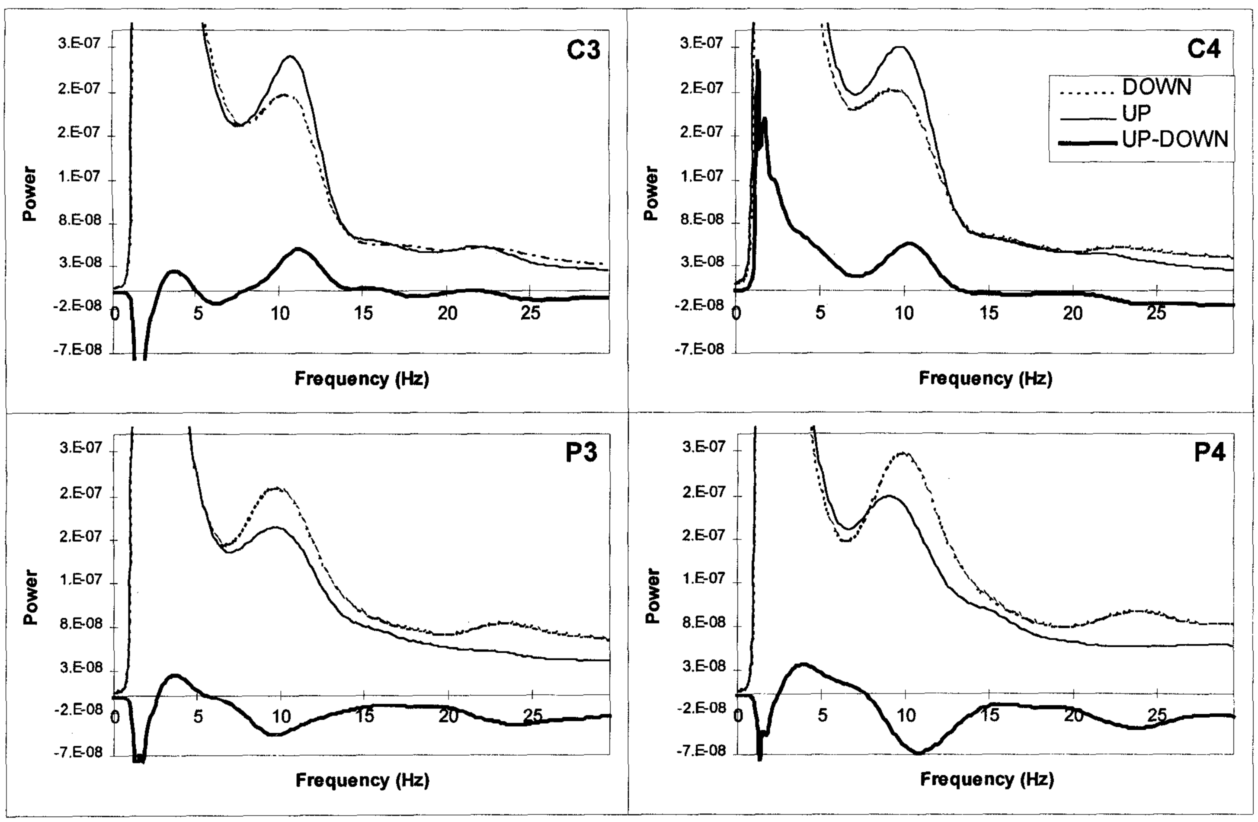

Figure 1. An example of FFT spectral difference between UP and DOWN EEG patterns recorded over central and parietal regions.

C. Off-Line Analysis

A half second window was used with a 0.45 second overlap to create the feature vectors. For the FFT method, the window was also passed through a Hanning window and a high pass filter with a 1 Hz cutoff to remove low frequency components. We extracted features for selected electrodes above the sensory-motor and parietal areas in both hemispheres (C3, C4, P3 and P4).

The FFT algorithm that was used is from LabWindows CVI (National Instruments) and the AR algorithm was taken with minor modifications from Press et. al. [14]. The correctness of these routines was cross-verified by the S-PLUS statistical software. The most common procedures for AR modeling are Kalman filtering, the Yule-Walker approach, and the Burg algorithm. We selected to use the Burg algorithm because it was shown by Jansen [13] to give superiour results in EEG analysis over the other two AR methods. A plot showing the averaged spectrum of the data used for testing can be seen in Fig. 1. This figure shows a significant difference between the up and down thought patterns in the μ-rhythm and in the higher β-rhythm. The averaged AR coefficients and their standard deviation can be seen in Fig. 2. As can be seen from Fig. 2, coefficients two and three seem to give most information for discriminating between the two patterns.

For comparing the information captured by each feature extraction method we looked at the probability of misclassification (PMC). PMC tells us the probability that each sample will be wrongly classified, and is a common and intuitive measure of performance for classification problems. ALN was the classifier that we used in both the on-line experiments and in this analysis. ALN is a non-linear adaptive system which is capable of learning any continuous function to any degree of accuracy [16].

III. RESULTS

The amount of data that was recorded for the training set was not equal for the up and down direction, therefore the training may have been slightly biased towards one of the classes. Results obtained by comparing classification performance using equal number of FFT spectral features and AR coefficients can be seen in Figure 3. This figure shows the difference in probability of misclassification between features extracted by FFT spectrum and AR coefficients. The FFT spectral features were equally divided between 2 and 30 Hz. Taking four FFT features per lead gives, as inputs to the classifier, the average power in 2-9 Hz, 9-16 Hz, 16-23 Hz, and 23-30 Hz frequency bands. We used up to 32 equally spaced frequency bands between 2 and 30 Hz which gives resolution of 0.875 Hz. At this resolution, the total number of input features to the classifier for all leads is 128, however the training set had 6014 samples, which should be large enough to generalize in a 128 dimensional input space.

As can be seen in Fig. 3, the optimal number of features per lead is four in both methods. With four features per lead, the analysis shows the AR coefficients to give the lower probability of misclassification.

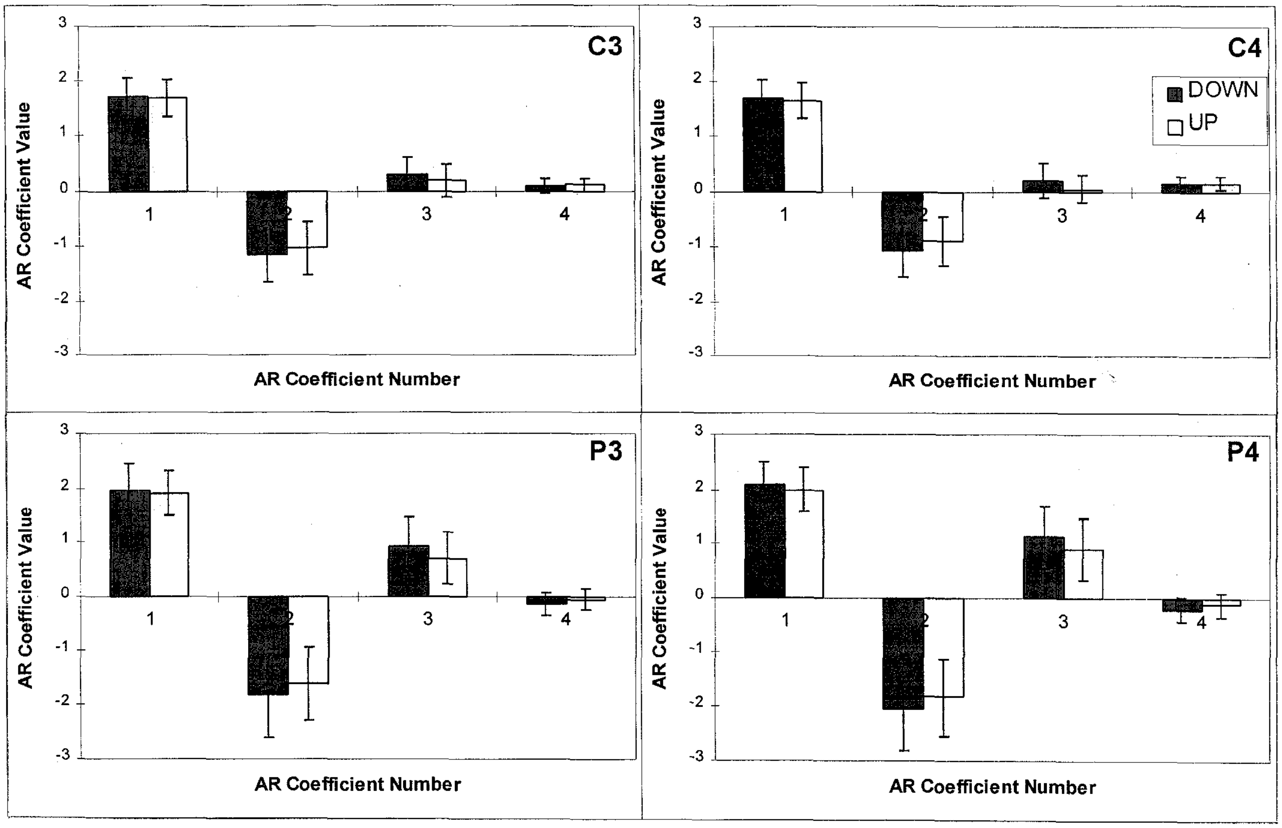

Figure 2. An example of the AR coefficients calculated using the same data from Fig. 1.

IV. DISCUSSION AND CONCLUSION

We trained a subject to control the movement of an object on a computer screen in up and down direction by having him use only consciously modulated EEG potentials. Off-line FFT frequency bands and raw AR coefficients were compared as methods for feature extraction. The AR method gave better performance, except at two features per lead.

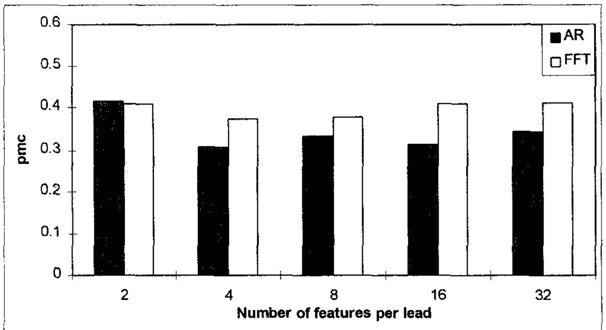

Figure 3 Probability of misclassification when using equal number of AR coefficients and FFT power bands .

Using spectral features may give better classification if the optimal frequency bands are selected for each subject. However, if only specific frequency bands are extracted from the data, then this hinders the subject in trying to find new ways to modulate their EEG. Not putting limitations on where in the frequency domain the features may be located is inherent when using the raw AR coefficients. AR coefficients seem to be the better alternative to FFT frequency bands for extracting features when no prior spectral information is assumed, however more tests should be done to make any definite conclusions.

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